MACHINE LEARNING-AUGMENTED SINGLE- AND MULTI-TRAIT SINGLE-STEP GBLUP ENHANCED GENOMIC PREDICTION IN WHITE SPRUCE+

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Background

Genomic prediction (GP) in forest tree breeding has traditionally relied on parametric linear models (GBLUP) assuming additive effects and linear genotype-phenotype relationships. These models may be suboptimal for complex traits influenced by dominance and epistasis, particularly in long-lived tree species where productivity, defense, and climate-related traits are often polygenic. Machine learning (ML) kernel methods, including Gaussian and arc-cosine kernels, offer a flexible alternative by implicitly capturing non-linear and interaction genetic effects, yet their integration into single-step GBLUP (ssGBLUP) is limited in forest trees.

Objectives

- Compare predictive ability of linear (VanRaden) *vs.* non-linear kernels (Gaussian, arc-cosine)
- Quantify gains in predictive ability for traits with complex (non-additive) genetic architecture
- Identify optimal kernels and trait responses
- Assess the contribution of additive, dominance, and epistatic effects to predictive performance

Materials and Methods

Data

- White spruce progeny test (Alberta, Canada)
- 30 traits (Table 1)
- Up to 3,936 trees (open-pollinated)
- 612 genotyped individuals
- 211,061 SNPs (GBS)

Models

- Single- and multi-trait ssGBLUP models
  - Linear kernel (G, VanRaden)
  - Gaussian kernel (GK, kernel averaging strategy)
  - Arc-cosine kernel (AK; 1-20 layers)
- Additional GBLUP model: Additive (A) + Dominance (D) + Epistasis (A×A, A×D)

Evaluation

- 10-fold cross-validation
- Predictive ability = Pearson correlation between predicted and reference breeding values

Table 1. Summary statistics and heritability estimates (h^2, and approximate standard errors) for the 30 traits assessed in the white spruce population based on raw phenotypes.

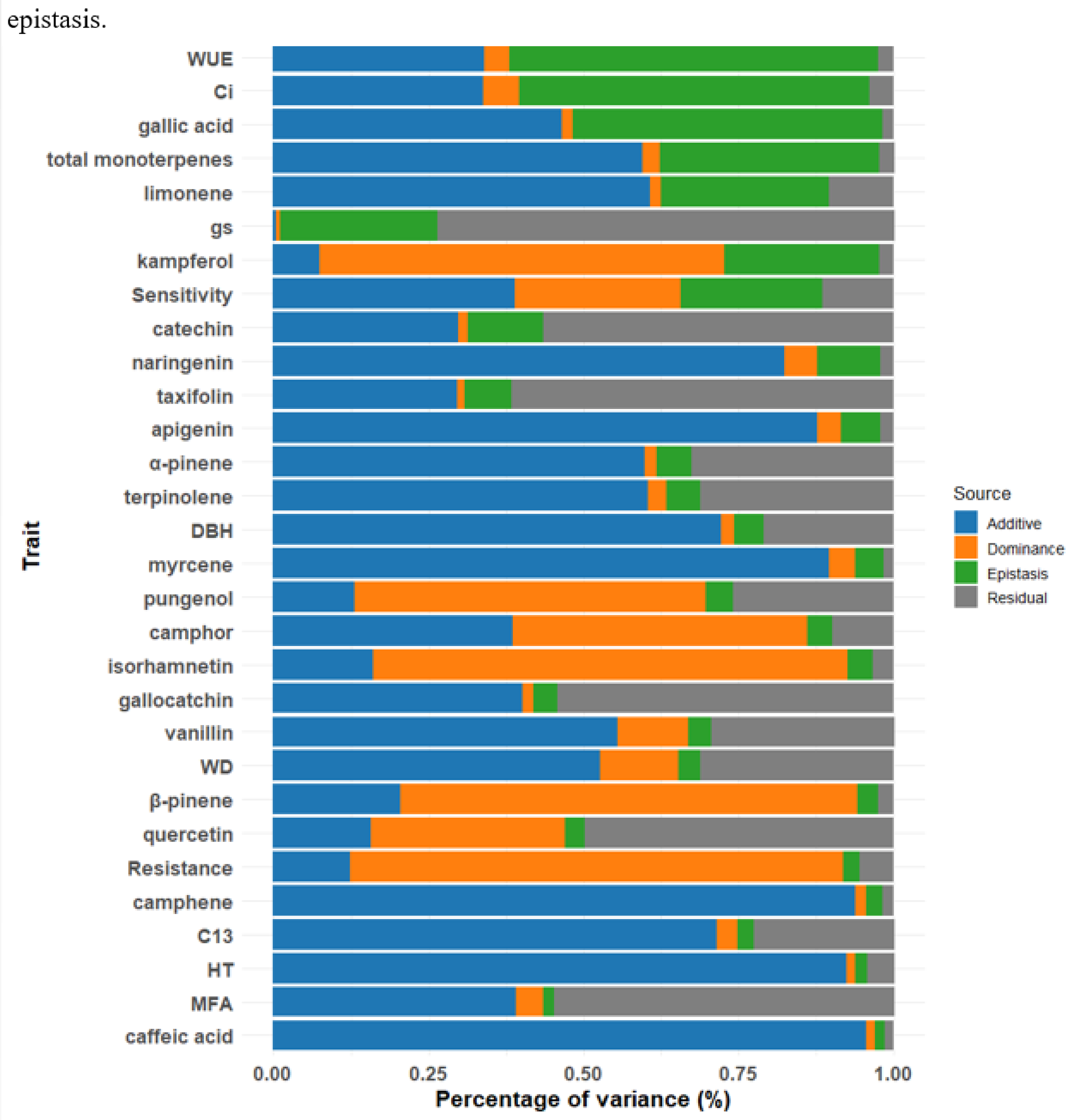
Type of trait	Trait full name	Trait	Unit	h^2	Mean	Min.	Max.
Productivity	Tree height	HT	cm	0.44 (0.08)	838.21	150	1290
	Diameter at breast height	DBH	cm	0.26 (0.10)	12.82	1	25
	Wood density	WD	$kg.m^{-3}$	0.42 (0.14)	379.60	304.07	497.64
	Microfibril angle	MFA[a]	°	0.35 (0.16)	21.27	17.15	50.32
Climate-adaptability	Drought resistance	Resistance	-	0.25 (0.18)	0.57	0.23	1.20
	Mean drought sensitivity	Sensitivity	-	0.33 (0.17)	0.21	0.03	0.38
	Average stable carbon isotope ratio	C13	-	0.58 (0.14)	-26.28	-28.14	-24.37
Defense chemical compounds	α-pinene	α-pinene[a]	$ng\ mg^{-1}$	0.50 (0.14)	221.32	13.99	1029.56
	β-pinene	β-pinene[a]	$ng\ mg^{-1}$	0.33 (0.19)	35.81	9.24	153.40
	camphene	camphene[a]	$ng\ mg^{-1}$	0.86 (0.14)	551.15	30.95	2585.76
	camphor	camphor[a]	$ng\ mg^{-1}$	0.29 (0.17)	629.99	30.23	3842.54
	myrcene	myrcene[a]	$ng\ mg^{-1}$	0.76 (0.14)	452.09	13.79	5644.61
	limonene	limonene[a]	$ng\ mg^{-1}$	0.49 (0.15)	606.88	13.42	3406.85
	terpinolene	terpinolene[a]	$ng\ mg^{-1}$	0.60 (0.18)	33.61	8.17	145.48
	total monoterpenes	total monoterpenes[a]	$ng\ mg^{-1}$	0.48 (0.15)	4037.35	33.44	15507.53
	gallic acid	gallic acid	mg/kg	0.23 (0.22)	0.80	0.23	1.55
	gallocatechi	gallocatchin[a]	mg/kg	0.35 (0.20)	154.34	14.22	487.61
	catechin	catechin	mg/kg	0.25 (0.22)	3332.53	721.07	7605.46
	pungenol	pungenol[a]	mg/kg	0.22 (0.23)	39.59	10.63	145.57
	caffeic acid	caffeic acid[a]	mg/kg	0.81 (0.17)	6.99	0.57	39.23
	vanillin	vanillin	mg/kg	0.51 (0.20)	7.42	2.88	13.46
	taxifolin	taxifolin[a]	mg/kg	0.50 (0.23)	20.93	7.41	80.90
	quercetin	quercetin[a]	mg/kg	0.10 (0.02)	16.67	3.39	70.60
	naringenin	naringenin[a]	mg/kg	0.56 (0.18)	2.17	0.48	38.54
	kaempferol	kaempferol[a]	mg/kg	0.18 (0.02)	122.37	11.20	507.22
	apigenin	apigenin[a]	mg/kg	0.52 (0.17)	0.50	0.14	3.13
	isorhamnetin	isorhamnetin[a]	mg/kg	0.19 (0.23)	1.13	0.20	6.08
Climate-adaptability	Stomatal conductance	gs	-	0.00 (0.02)	0.17	0.02	0.58
	Intrinsic water use efficiency	WUE	-	0.26 (0.25)	0.08	0.03	0.19
	Intercellular CO_2 concentration	Ci	-	0.26 (0.25)	242.50	66.61	322.25

NOTE: [a] Logarithmic transformed for heritability calculation, mean (Mean), standard deviation (SD), minimum (Min.), and maximum (Max.) values.

Results

Genetic variance partitioning across traits

Figure 1. Partitioning of variance components for each trait based on the GBLUP model using a standard linear kernel (VanRaden's genomic relationship matrix, G). Traits are ordered from highest to lowest according to the proportion of total genetic variance attributed to epistasis.

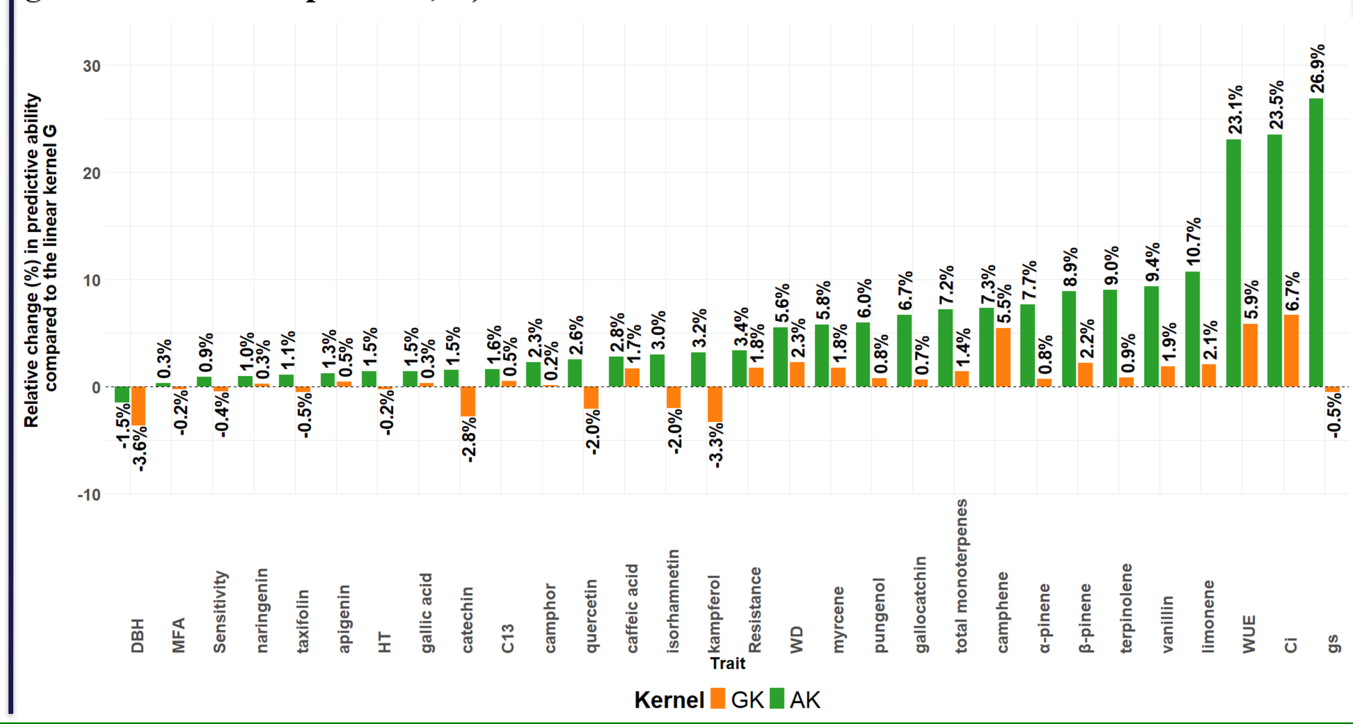


Predictive ability using single-trait ssGBLUP

Figure 2. Average predictive ability for the 30 studied traits using single-step GBLUP models with three genomic relationship kernels: the standard linear kernel (VanRaden's, G), and two non-linear kernels (the Gaussian kernel -GK- and the deep arc-cosine kernel -AK-). Average predictive abilities are shown below each boxplot.

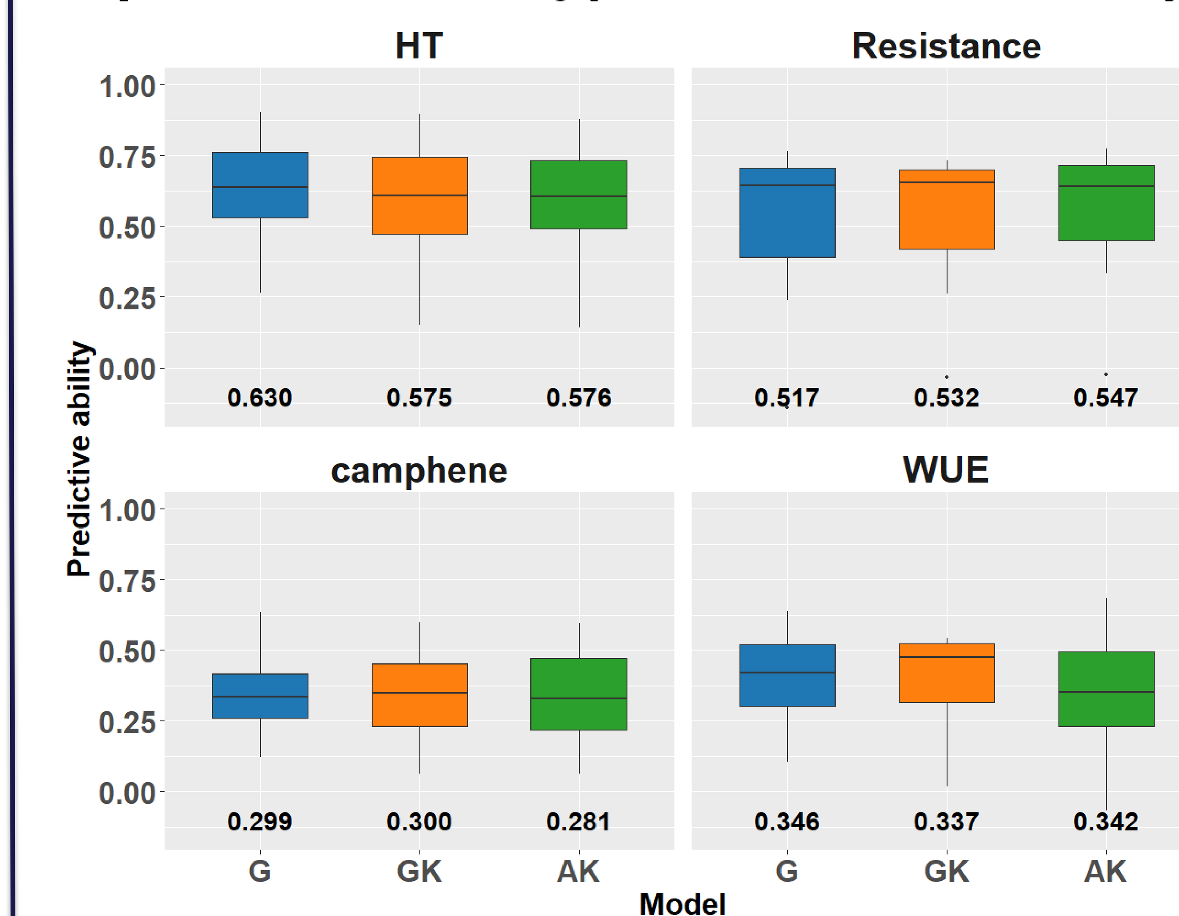
Relative gains in predictive ability

Figure 3. Relative change in predictive ability (%) for the non-linear Gaussian kernel (GK) and deep arc-cosine kernel (AK) compared to the standard linear kernel (VanRaden's genomic relationship matrix, G).



Predictive ability using multi-trait ssGBLUP

Figure 4. Average predictive ability for the four representative studied traits using multi-trait single-step GBLUP models with three genomic relationship kernels: the standard linear kernel (VanRaden's, G), and two non-linear kernels (the Gaussian kernel -GK- and the deep arc-cosine kernel -AK-). Average predictive abilities are shown below each boxplot.



Conclusions

1. Non-linear kernels improved genomic prediction across most traits in white spruce, with strong trait-specific effects
2. The AK consistently achieved the highest predictive ability, particularly for traits with complex non-additive genetic architectures
3. The GK showed moderate gains, while the linear model remained competitive for productivity traits (e.g., DBH)
4. In multi-trait analyses, predictive ability was similar across kernels; improvements were mainly driven by cross-trait information, with notable gains for low-heritability traits (e.g., Resistance)
5. Kernel choice strongly influences prediction accuracy and should be tailored to trait genetic architecture
6. Integrating ML kernels can enhance prediction of complex traits, supporting accelerated genetic gain and climate-resilient breeding

Take-home message

Aligning kernel complexity with genetic architecture improves prediction and enhances selection efficiency, particularly with AK

Sponsors

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